A Lightweight Spatial–Spectral Deformable CNN for UAV Hyperspectral Image Classification

Xiaohu Ma, Mengmeng Zhang, Zheng Kan, Yunhao Gao, and Wei Li, *Senior Member, IEEE*

***Abstract*—In recent years, uncrewed aerial vehicle (UAV) technology has shown great potential for application in hyperspectral image (HSI) classification tasks due to its advantages of flexible scheduling and fast response. However, existing deep learning-based classification algorithms have not thoroughly studied the problems caused by the higher spatial resolution of UAV HSIs. Such as more severe intraclass variation and higher computational resource consumption. These issues limit classification performance. To address these challenges, this article proposes a lightweight spatial–spectral deformable convolutional neural network (LS^{2}DCNet) for UAV HSI classification. This network reduces computational resource usage and data processing time, meeting the fast-response application requirements while ensuring classification performance. First, spatial–spectral deformable convolution (S^{2}DConv) is designed to construct a lightweight feature extraction network. This not only enhances the adaptive extraction of fine features but also reduces computational resource consumption and improves response speed. Second, a dynamic labeling-based joint loss (DL-JLoss) is designed to dynamically learn the distribution relationship between data classes. This improves the network's generalization performance. A more realistic experimental validation was conducted on three UAV HSI datasets using regionally divided training samples, and inference comparisons were performed on embedded devices. The results show that the proposed LS^{2}DCNet exhibits a better overall performance in terms of classification accuracy and inference speed. The relevant code can be found at https://github.com/niuroushu/A-Lightweight-Spatial-Spectral-Deformable-CNN-for-UAV-Hyperspectral-Image-Classification**

***Index Terms*—Hyperspectral image (HSI) classification, lightweight, spatial–spectral deformable convolution (S^{2}DConv), uncrewed aerial vehicle (UAV).**

I. Introduction

HYPERSPECTRAL images (HSIs) contain spatial information as well as spectral information from visible light to shortwave infrared, which can help distinguish the properties of objects. Therefore, HSI-based pixel-by-pixel classification tasks have been widely studied, such as land cover classification [1], [2], tree species classification [3], crop disease detection [4], and medical disease detection [5].

With the development of artificial intelligence technology, current HSI classification algorithms are mainly constructed based on deep learning. Hong et al. [6] proposed the SpectralFormer network based on the Transformer architecture to enhance the contextual modeling capabilities of spectral features in HSI. Wang et al. [7] introduced the capsule attention network (CAN), which leverages activity vectors and attention mechanisms to improve HSI classification. Wang et al. [8] developed a dual stochastic graph learning method (HPCDL) based on graph convolutional networks (GCNs), which captures useful spectral information by learning projection spaces for large-scale HSI projection clustering. Li et al. [9] presented a novel HSI classification model, MambaHSI, based on the Mamba model, capable of modeling long-range interactions across the entire image while adaptively integrating spatial and spectral information. Among various deep learning models, convolutional neural network (CNN) have been widely applied in HSI classification due to their simple and effective architecture. Hu et al. [10] used CNNs to directly extract spectral features of HSI for classification and obtained better performance than some conventional methods. However, HSI has spectral variability, and using only spectral information for classification makes it difficult to classify certain pixels correctly. Therefore, the joint utilization of spatial and spectral information of HSI has gradually become the main research direction. Zhong et al. [11] proposed an end-to-end spectral–spatial residual network (SSRN) for HSI classification, where residual blocks constructed based on 3-D convolution are successively used to extract discriminative spatial–spectral features. Wan et al. [12] designed an efficient spectral–spatial fusion method (ESSF) for HSI classification, which combines a multiscale feature fusion module as well as an adaptive attention mechanism to further improve the classification accuracy. Deformable convolution (DConv) [13], [14], [15], [16], [17] is a variant of convolution that enhances the convolutional kernel's ability to adapt to geometric variations of objects by introducing learnable spatial offsets. It has achieved satisfactory results in the field of image classification. Therefore, in the field of HSI classification, Zhao et al. [18] directly introduced Deformable ConvNets v2 (DCNv2) to construct a hybrid HSI classification network, enhancing spatial feature representation and improving classification performance. Wang et al. [19] incorporated deformable convolution residual modules to strengthen the network's adaptability to

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Xiaohu Ma, Mengmeng Zhang, Yunhao Gao, and Wei Li are with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: 2441129931@qq.com; mengmengzhang@bit.edu.cn; gaoyunhao@bit.edu.cn liw@bit.edu.cn;).

Zheng Kan is with Beijing University of Technology, Beijing 100021, China (e-mail: kanzheng05@163.com).

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various complex shapes of ground objects, further learning complex spatial information. Other HSI classification networks such as [20], [21], [22], [23], and [24] have also achieved good results.

However, as the model complexity continues to increase, deep networks gradually face the problem of excessive computational resource consumption. This is particularly prominent on edge devices with limited computational power. To balance the model performance and deployment cost, lightweight HSI classification algorithm design has become a new research hotspot. Cui et al. [25] designed a lightweight network (LiteDepthwiseNet) for HSI classification based on 3-D depth-separable convolution, which achieves advanced performance with fewer parameters and computational costs. Cui et al. [26] proposed a lightweight spectral–spatial attention network (LSSAN) for HSI classification based on MobileNetV3. A more concise and efficient spatial attention module is designed to reduce the redundant calculation problem and improve classification accuracy. Ma et al. [27] proposed a hybrid convolution-based lightweight network (Lite-HCNet) for HSI classification. A lighter and more efficient DSC (LE-DSC) is designed to further reduce the size and improve the performance of conventional depth-separable convolution. Other lightweight HSI classification algorithms, such as [28], [29], [30], [31], and [32], have also achieved good results in reducing the consumption of computing resources.

Recently, with the rapid development of uncrewed aerial vehicle (UAV) technology and its widespread use in emergency remote sensing scenarios, existing algorithms face new adaptability challenges. Compared to traditional satellite platforms, UAVs offer the advantages of quick response and flexible deployment. The HSI captured by UAVs show dual characteristics below.

1) UAV HSI offers higher spatial resolution, typically reaching the submeter or even centimeter levels, which significantly enhances the ability to represent detailed ground objects. However, the increase in resolution not only enriches the information content but also introduces more complex shapes of ground objects and more pronounced intraclass spectral variability, thereby increasing the difficulty of classification [33].
2) Regarding application requirements, satellite HSI typically emphasizes large-scale, periodic monitoring, where data processing can depend on ample computational resources and extended timeframes; in contrast, UAV HSI is often employed in emergency response and other situations that demand quick decision-making, requiring more efficient and precise algorithms within limited computational resources.

Facing the unique properties of UAV HSI, the existing algorithms have certain limitations. Its lightweight algorithm still has room for optimization in maintaining the balance between accuracy and efficiency. In addition, the existing algorithms have not thoroughly studied how to enhance the adaptive extraction of fine features of UAV HSI and improve the generalization performance of the data.

To address the above challenging problems, this article proposes a lightweight spatial–spectral deformable convolutional neural network (LS^{2}DCNet) for UAV HSI classification. LS^{2}DCNet consists of spatial–spectral deformable convolution (S^{2}DConv) and dynamic labeling-based joint loss (DL-JLoss). S^{2}DConv is used to construct a lightweight feature extraction network for self-adaptive extraction of fine-grained features of UAV HSI. DL-JLoss is developed to improve the generalization performance of the network by learning the interclass distribution relationships of ground objects. The specific contributions are summarized as follows.

1) Through the codesign of spatial deformable convolution (Spatial-DConv) and spectral deformable convolution (Spectral-DConv), S^{2}DConv enables adaptive extraction of finer spatial and spectral features from high-resolution UAV HSI data, enhancing discriminative power for ground objects.
2) By developing a dynamic soft label generation mechanism with interclass distribution constraints, DL-JLoss jointly optimizes adaptive soft labels and hard labels. This effectively strengthens the network's ability to model interclass relationships among ground objects and improves generalization performance.
3) Constructed the lightweight network framework LS^{2}DCNet. Leveraging the collaborative optimization of S^{2}DConv and DL-JLoss, LS^{2}DCNet maintains high classification accuracy while reducing model computational complexity, achieving an optimal balance between efficiency and accuracy.

The rest of this article is organized as follows. In Section II, the proposed method is introduced in detail, including the whole network structure and the local module structure. Then, in Section III, a series of experiments and analyses are carried out to verify the effectiveness of the method. Finally, the thesis is summarized in Section IV.

II. Methods

In this section, the overall overview of the proposed LS^{2}DCNet is first introduced. Then S^{2}DConv and DL-JLoss were introduced separately.

A. Lightweight Spatial–Spectral Deformable Convolutional Neural Network

The CNN-based HSI pixel-by-pixel classification algorithm follows the process of extracting neighborhood blocks for each HSI pixel centered on it, and the output data is represented as $\boldsymbol{X} \in R^{D\times D\times C}$, where $\boldsymbol{D}$ is the width and height of the spatial dimension and $\boldsymbol{C}$ is the number of spectral bands. $\boldsymbol{X}$ sends the semantic information representing its central pixel into the network for classification. LS^{2}DCNet also follows this process. In addition, based on the guiding principle of lightweight and fast processing, after data preprocessing, the data fed into LS^{2}DCNet is represented as $\boldsymbol{X} \in R^{3\times 3\times C}$, with a spatial dimension set to 3×3 to reduce the computational complexity of data processing, and a spectral dimension set to the original number of bands $\boldsymbol{C}$ to preserve rich spectral information. The internal structure of LS^{2}DCNet is shown in Fig.1, which includes two parts: a lightweight feature extraction network based on S^{2}DConv and a lightweight classification head based on center-guided global average pooling (CG-GAP).

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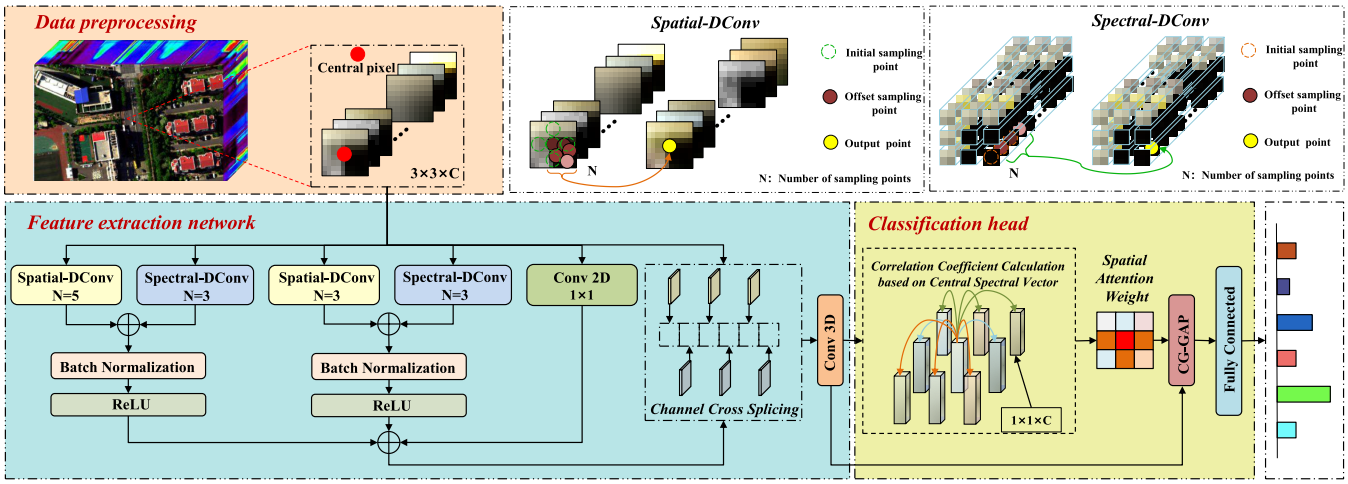


Fig. 1. Detailed framework diagram of LS^{2}DCNet.

1) Feature Extraction Network: The feature extraction network of LS^{2}DCNet is designed as a multibranch parallel structure for multiview spatial, spectral feature extraction, and fusion of input $\boldsymbol{X}$. This includes two spatial and spectral feature fusion branches, one feature linear variation branch, and one original feature branch.

Two spatial and spectral feature fusion branches are implemented using S^{2}DConv with two different convolution kernel sizes, which are used to adaptively extract multiscale spatial–spectral joint features of UAV HSI and enhance the expression ability of fine features. The operation flow of S^{2}Dconv can be represented as a function F_1 (.)

$$F_1(\boldsymbol{x}) = \sigma\left(f_{\mathrm{BN}}\left(f_{\mathrm{Spa}}(\boldsymbol{x}) + f_{\mathrm{Spe}}(\boldsymbol{x})\right)\right) \tag{1}$$

where $\boldsymbol{x}$ is the input data; f_{Spa} (.) is Spatial-DConv, used for adaptively extracting 2-D spatial features of UAV HSI; f_{Spe} (.) is Spectral-DConv, used for adaptively extracting 1-D spectral features of UAV HSI; σ (.) is the ReLu activation function used to enhance the nonlinear representation of features; F_{BN} (.) is BN used to accelerate network convergence.

The feature linear change branch is implemented by a fully grouped 2-D convolution with a kernel size of 1×1, enriching feature representation through simple linear changes with minimal computational complexity. The characteristic linear change can be expressed as a function F_2 (.)

$$F_2\left(\boldsymbol{x}_{i,j,c}\right) = \boldsymbol{x}_{i,j,c} \cdot \boldsymbol{W}_{i,j,c} + \boldsymbol{b} \tag{2}$$

where $\boldsymbol{x}_{i,j,c}$ is the values of input data $\boldsymbol{x}$ at spatial position (i, j) and channel position $\boldsymbol{c}$, $\boldsymbol{W}_{i,j,c}$ is the corresponding position weights, and $\boldsymbol{b}$ is the bias parameter.

After the output features of the three feature extraction branches are added and fused, they are represented as $\boldsymbol{X}^1$. Then, $\boldsymbol{X}^1$ is weighted and fused with the original feature $\boldsymbol{X}$ to output $\boldsymbol{X}^2$, and the weights are generated by 3-D convolution. Specifically, after cross-stitching between channels, $\boldsymbol{X}^1$ and $\boldsymbol{X}$ are merged pixel by pixel using a grouped 3-D convolution with a kernel size of $2 \times 1 \times 1$, a step size of $(2, 1, 1)$, and a number of groups $\boldsymbol{C}$. The weighted fusion with the original features can be expressed as a function F_3 (.)

$$F_3\left(\boldsymbol{x}^1_{i,j,c}, \boldsymbol{x}_{i,j,c}\right) = \boldsymbol{x}^1_{i,j,c} \cdot \boldsymbol{W}^1_{i,j,c} + \boldsymbol{x}_{i,j,c} \cdot \boldsymbol{W}_{i,j,c} + \boldsymbol{b} \tag{3}$$

where $\boldsymbol{x}^1_{i,j,c}$ and $\boldsymbol{x}_{i,j,c}$ are the values of $\boldsymbol{X}^1$ and $\boldsymbol{X}$ at spatial position (i, j) and channel position $\boldsymbol{c}$, respectively. $\boldsymbol{W}^1_{i,j,c}$ and $\boldsymbol{W}_{i,j,c}$ are the corresponding position weights, and $\boldsymbol{b}$ is the bias parameter.

2) Classification Head: Following the guiding principle of lightweight network structure, LS^{2}DCNet adopts a simple classification head structure, namely a single-layer global average pooling (GAP) layer and a single-layer fully connected layer (FC). To further reduce computational complexity, this article designs CG-GAP to replace the standard GAP. The calculation flow of CG-GAP is shown in Fig. 1.

Unlike standard GAP, CG-GAP does not average pool pixels across the entire spatial dimension but selects $\boldsymbol{N}$ points for average pooling. These $\boldsymbol{N}$ points are sorted and selected based on the attention weight of each pixel in the spatial dimension. It is recommended that $\boldsymbol{N}$ be set to half the number of spatial pixels to balance accuracy and efficiency. In CG-GAP, the method of generating spatial attention weights is specifically designed for HSI. Because the overall label of the input data is the label of the central pixel, the correlation coefficient is calculated based on the spectral vector of the spatial center position and the spectral vectors of the remaining positions, and the resulting value is the spatial attention weight of the corresponding position.

B. Spatial–Spectral Deformable Convolution

DConv enhances the adaptability of convolution kernels to object geometric deformations by introducing learnable spatial offsets, enabling better capture of subtle changes in object features. This approach is particularly well-suited for high-resolution UAV HSI. While previous work [18], [19] has introduced DConv into HSI classification, it has only employed standard DConv in a straightforward manner. Standard DConv focuses solely on deformation in a single spatial

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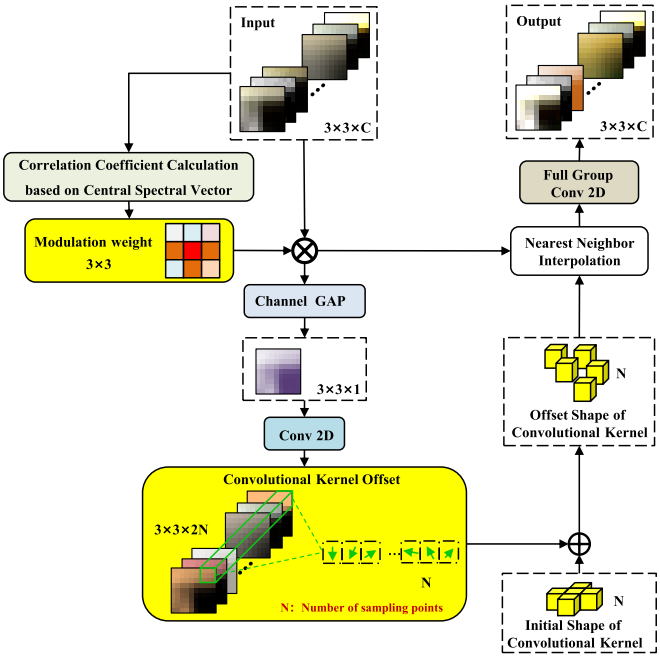


Fig. 2. Detailed framework diagram of Spatial-DConv.

dimension and cannot effectively focus on the rich and critical spectral features in UAV HSI, resulting in limitations in spectral information extraction. To address the dual requirements of complex spatial structure and rich spectral information in UAV HSI data, this article specifically designs S2DConv for adaptive extraction of spatial–spectral joint features in UAV HSI. The proposed S2DConv is based on a specific data transformation structure, cleverly combining Spatial-DConv and Spectral-DConv. The Spatial-DConv module adapts the convolution kernel sampling grid to focus on key spatial regions. The Spectral-DConv module learns to select discriminative spectral bands through spectral offset learning. The combination of the two not only compensates for the limitations of single-dimensional transformations but also achieves deep integration of spatial and spectral features, enhancing the richness and adaptability of feature representation.

1) Spatial-DConv: Spatial-DConv is designed to extract spatial features of UAV HSI. The detailed structure is shown in Fig. 2, which includes three main parts: modulation weight generation, offset generation, and offset convolution calculation, which together achieve efficient and accurate spatial feature extraction. The operation function of Spatial-DConv can be expressed as $f_{\text{Spa}}(.)$

$$f_{\text{Spa}}\left(\boldsymbol{x}_{p_0}\right) = \sum_{n=1}^{N} \boldsymbol{W}_n \cdot \boldsymbol{x}_{(p_0+p_n+\Delta p_n)} \cdot \boldsymbol{m}_n + \boldsymbol{b} \tag{4}$$

where $\boldsymbol{x}_{p_0}$ refers to the value of the p_0 position in the feature map, $\boldsymbol{N}$ is the number of sampling points in the convolutional kernel, $\boldsymbol{W}_n$ is the weight of the convolutional kernel corresponding to the sampling point, $\boldsymbol{m}_n$ is the feature modulation weight corresponding to the sampling point, p_n is the sampling position of the convolutional kernel, Δp_n is the offset of the p_n position, and $\boldsymbol{b}$ is the bias parameter.

Feature modulation weights in standard DConv are used to constrain the weights of individual feature points during the offset convolution operation, thereby improving stability. Spatial-DConv calculates feature modulation weights through a spatial attention mechanism. These weights not only replace the modulation weights generated by additional 2-D convolutions in standard DConv, reducing computational complexity, but also accurately reflect the spatial focus of UAV HSI data, ensuring that feature point weights are properly adjusted during subsequent offset convolution processes. This enhances the network's stability and expressive capacity. The calculation process for modulation weights aligns with CG-GAP, effectively aggregating spatial information.

The convolution kernel offset is generated through an additional 2-D convolution in standard DConv, but in this process, the input channel is the channel of the original data, which can result in a significant amount of computation. Therefore, to reduce computational complexity, Spatial-DConv uses channel GAP compression to compress channels, followed by convolution operations to generate convolution kernel offsets. The number of channels for the offset of the convolution kernel is $\boldsymbol{2N}$, representing the horizontal and vertical coordinates of the $\boldsymbol{N}$ sampling points of the convolution kernel. The offset generation depends on the spatial region emphasized by the modulation weight, and the two complement each other in the spatial dimension. The offset adjusts the sampling point position, while the modulation weight influences the contribution of the sampling point. Together, they enable more flexible and accurate spatial feature capture.

The convolution kernel shape of standard DConv is regular. Usually $K \times K$, with a square growth in the number of parameters, which limits the trade-off between network overhead and performance. Therefore, inspired by the LDConv [17]. The initial convolution kernels of arbitrary shapes are designed in the Spatial-DConv to balance network overhead and performance. For example, in Fig. 2, the offset convolution calculation part uses an initial convolution kernel with a cross shape and five sampling points. The initial convolution kernel is combined with the convolution kernel offset to obtain the offset convolution kernel shape.

In the final part of Spatial-DConv, the interpolation calculation and resampling of the data are performed at each sample point of the offset convolution kernel. Subsequently, the output is computed by fully grouped 2-D convolution with a convolution kernel size $N \times 1$ for the $\boldsymbol{N}$ resampled points. The interpolation calculation of Spatial-DConv uses nearest-neighbor interpolation instead of bilinear interpolation in standard DConv. This is because the spatial size of the input data in this method is designed to be small at 3×3, so even the nearest interpolation can cover most of the area without bilinear interpolation.

In Spatial-DConv, modulation weights emphasize key feature regions using a spatial attention mechanism. Offset generation lowers computational load and flexibly adjusts sampling point positions through channel compression. Offset convolution combines these approaches to enable efficient spatial feature extraction via dynamic convolution kernels and interpolation resampling. The combined effect of these three

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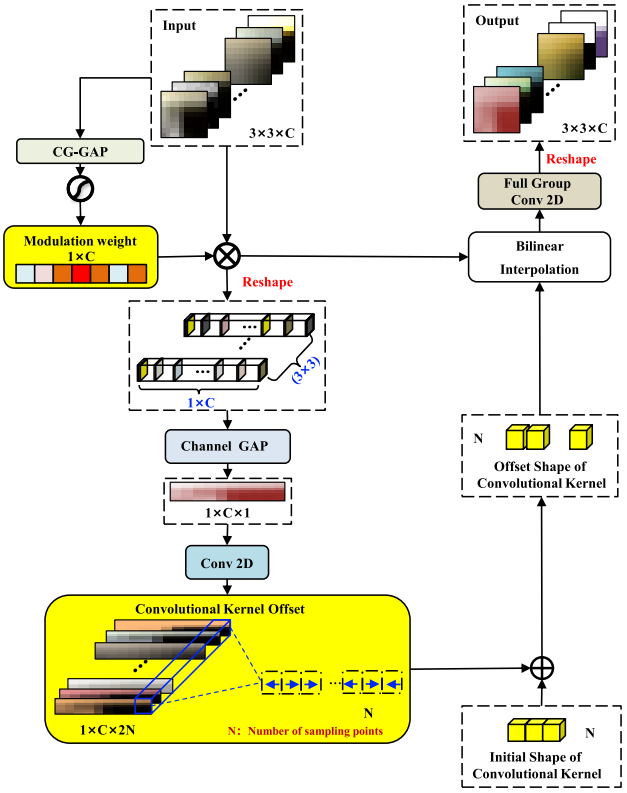


Fig. 3. Detailed framework diagram of Spectral-DConv.

methods significantly enhances the expression of UAV HSI spatial features while optimizing computational efficiency and performance.

2) Spectral-DConv: Spectral-DConv is designed to extract spectral features of UAV HSI. The detailed structure is shown in Fig. 3, which also includes three main parts: modulation weight generation, offset generation, and offset convolution calculation. The principle of Spectral-DConv is basically the same as Spatial-DConv, but the difference is that Spectral-DConv performs specific structural transformations on data. The channel dimension of the data is transformed into a spatial dimension, and the original spatial dimension is stretched and transformed into a channel dimension. Through this specific structural transformation, offset convolution calculations are achieved in the original spectral dimension.

In addition, since Spectral-DConv focuses on the spectral characteristics of UAV HSI, spectral attention weights are used in the modulation weight generation part of Spectral-DConv. Spectral attention weights are generated using a sigmoid after CG-GAP calculation on input data.

It is worth noting that Spectral-DConv uses bilinear interpolation instead of nearest neighbor interpolation. This is because UAV HSI has rich spectral information and requires a calculation method with a larger coverage area.

C. DL-JLoss

The core idea of label smoothing (LS) [34] is to generate a new type of label. It reduces the value of the target class

Algorithm 1 Dynamic Labeling-Based Joint Loss

Input: Predictions $q \in \mathbb{R}^{N\times C}$, labels $y \in \mathbb{Z}^N$, smoothing $\varepsilon \in (0,1)$, temperature T, balance $\alpha \in [0,1]$, stability δ

Output: Loss $\mathscr{L}$

$pred \leftarrow \arg\max(q, \text{dim} = 1)$
$mask \leftarrow (pred == y)$
Initialize $K \in \mathbb{R}^{N\times C}$ with zeros
$K[mask] \leftarrow \text{Softmax}(q[mask])$
$K[\neg mask] \leftarrow \text{Softmax}(q[\neg mask]/T)$
$S \leftarrow \varepsilon \times K$
$S[\text{range}(N), y] \leftarrow 1 - \varepsilon + \varepsilon \times K[\text{range}(N), y]$
Initialize one-hot $H \in \mathbb{R}^{N\times C}$ with zeros
$H[\text{range}(N), y] \leftarrow 1$
$\mathscr{L}_1 \leftarrow \text{KL_div}(\log(q+\delta), S+\delta, \text{reduction} =' batchmean')$
$\mathscr{L}_2 \leftarrow \text{CrossEntropyLoss}(q, H)$
$\mathscr{L} \leftarrow \alpha\mathscr{L}_1 + (1-\alpha)\mathscr{L}_2$
return $\mathscr{L}$

in the labels while increasing the value of the nontarget class, thus reducing the network's overconfident predictions during the training process. LS enhances the network's tolerance to different classes and improves the generalization ability. However, setting the same value for nontarget classes in the new labels generated by standard LS ignores the distribution relationship between different categories, which may bring less or even negative benefits in UAV HSI with complex category relationships. Therefore, this article specifically designs DL-JLoss for the training process of LS^{2}DCNet, which enhances the network's generalization performance and improves classification accuracy by generating dynamic soft labels with interclass distribution relationships and combining them with original hard labels for joint loss calculation. The algorithm flow of DL-Jloss is shown in Algorithm 1.

The core of DL-JLoss is to generate dynamic soft labels with interclass distribution relationships. Specifically, in each cycle of network training, the current prediction results $\boldsymbol{q}(\boldsymbol{x}_i)$ of the network are used to generate the interclass distribution relationship $\boldsymbol{K}_i$. Among them, the correct results predicted by the network are considered reliable distributions, and the distribution relationship weights of each category are directly obtained through softmax normalization. The results of network prediction errors are considered as an unreliable distribution, and the temperature parameter $\boldsymbol{T}$ needs to be set in softmax normalization to flatten its distribution and reduce the impact of incorrect results. Using the current prediction results to generate $\boldsymbol{K}_i$ can reduce computational storage and ensure the timeliness of network learning. The calculation process of the interclass distribution relationship $\boldsymbol{K}_i$ is as follows:

$$\boldsymbol{K}_i = \begin{cases} \text{Softmax}\left(\boldsymbol{q}\left(\boldsymbol{x}_i\right)\right), & \boldsymbol{q}\left(\boldsymbol{y}_i \middle| \boldsymbol{x}_i\right) = \text{Max}\left(\boldsymbol{q}\left(\boldsymbol{x}_i\right)\right) \\ \text{Softmax}\left(\boldsymbol{q}\left(\boldsymbol{x}_i\right)/\boldsymbol{T}\right), & \boldsymbol{q}\left(\boldsymbol{y}_i \middle| \boldsymbol{x}_i\right) \neq \text{Max}\left(\boldsymbol{q}\left(\boldsymbol{x}_i\right)\right). \end{cases} \tag{5}$$

After obtaining the interclass distribution relationship $\boldsymbol{K}_i$, the next step is to generate dynamic soft labels. The calculation

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TABLE I
DETAILED CLASS INFORMATION AND THE NUMBER OF TRAINING AND TESTING SAMPLES ON THE EXPERIMENTAL DATASET

No.	WHU-Hi-LongKou Class Name	Training	Testing	QUH-Qingyun Class Name	Training	Testing	QUH-Tangdaowan Class Name	Training	Testing
1	Corn	100	34411	Trees	500	277650	Rubber track	100	25749
2	Cotton	100	8274	Concrete building	500	179012	Flaggingv	100	55453
3	Sesame	100	2931	Car	500	13283	Sandy	100	33937
4	Broad-leaf soybean	100	63112	Ironhide building	500	9267	Asphalt	100	60590
5	Narrow-leaf soybean	100	4051	Plastic playground	500	217236	Boardwalk	100	1762
6	Rice	100	11754	Asphalt road	500	255646	Rocky shallows	100	37025
7	Water	100	66956				Grassland	100	14027
8	Roads and houses	100	7024				Bulrush	100	63987
9	Mixed weed	100	5129				Gravel road	100	30595
10							Ligustrum vicaryi	100	1683
11							Coniferous pine	100	21136
12							Spiraea	100	649
13							Bare soil	100	1586
14							Buxus sinica	100	786
15							Photinia serrulata	100	13920
16							Populus	100	140804
17							Ulmus pumila L	100	9702
18							Seawater	100	42175
	TOTAL	900	203642	TOTAL	3000	951893	TOTAL	1800	555566

process of dynamic soft label $\boldsymbol{S}_{y_i}$ is as follows:

$$\boldsymbol{S}_{y_i} = \begin{cases} \varepsilon \cdot \boldsymbol{K}_i, & \boldsymbol{i} \neq \text{target} \\ (1-\varepsilon) + \varepsilon \cdot \boldsymbol{K}_i, & \boldsymbol{i} = \text{target}. \end{cases} \tag{6}$$

The value range of ε is $(0, 1)$, used to control the overall size of nontarget values in the new label. The dynamic soft labels $\boldsymbol{S}_{y_i}$ with interclass distribution relationships are used to guide network training, avoiding excessive determinism caused by the network only learning the original hard labels. The representation of the original hard label $\boldsymbol{H}_{y_i}$ is as follows:

$$\boldsymbol{H}_{y_i} = \begin{cases} 0, & \boldsymbol{i} \neq \text{target} \\ 1, & \boldsymbol{i} = \text{target}. \end{cases} \tag{7}$$

The calculation process of the final total loss is as follows:

$$\begin{aligned} \boldsymbol{L} &= \alpha \cdot \boldsymbol{L}_1\left(\boldsymbol{S}_{y_i}, \boldsymbol{q}\left(\boldsymbol{x}_i\right)\right) + (1-\alpha) \cdot \boldsymbol{L}_2\left(\boldsymbol{H}_{y_i}, \boldsymbol{q}\left(\boldsymbol{x}_i\right)\right) \\ &= \alpha \cdot \sum_{j=1}^{n} \left(\boldsymbol{S}_{y_i} + \delta\right) \log\left(\frac{\boldsymbol{S}_{y_i} + \delta}{\boldsymbol{q}\left(j \middle| \boldsymbol{x}_i\right)} + \delta\right) \\ &\quad - (1-\alpha) \cdot \log\left(\boldsymbol{q}\left(j = \boldsymbol{y}_i \middle| \boldsymbol{x}_i\right)\right). \end{aligned} \tag{8}$$

Among them, $\boldsymbol{L}_1$ is the KL divergence loss, which is used to calculate the similarity between the probability distributions of the dynamic soft label $\boldsymbol{S}_{y_i}$ and the network prediction result $\boldsymbol{q}(\boldsymbol{x}_i)$. Among them, δ is the smoothing factor, usually set to a very small positive number, used to avoid zero probability problems and improve the numerical stability and robustness of the network. $\boldsymbol{L}_2$ is the cross-entropy loss (CEL) used to calculate the similarity between the probability distributions of the original hard label $\boldsymbol{H}_{y_i}$ and the network prediction result $\boldsymbol{q}(\boldsymbol{x}_i)$. α is used to balance $\boldsymbol{L}_1$ and $\boldsymbol{L}_2$.

III. EXPERIMENTS AND ANALYSIS

In this section, the three UAV hyperspectral datasets used in the experiment are first introduced, then the detailed network parameter settings are introduced, and finally, the experimental results are shown and analyzed.

A. Datasets Description

The three UAV hyperspectral datasets include the WHU-Hi-LongKou (LK) dataset, the QUH-Qingyun (QY) dataset, and the QUH-Tangdaowan (TDW) dataset.

The LK dataset was collected from 13:49 to 14:37 on 17 July 2018 using a DJI M600 Pro drone mounted with Nano-Hyperspec in Longkou, Hubei, China. The flight height of the UAV is 500 m, the space size of the dataset is 550×400 pixels, there are 270 spectral bands from 400 to 1000 nm, and the spatial resolution is about 0.463 m, including nine types of ground objects.

The QY dataset was collected from 14:06 to 14:45 on May 18, 2021, using a DJI M600 Pro drone mounted with Gaiaskymini2-VN near the primary school and residential area of Qingyun Road, Qingdao, China. The flight height of the UAV is 300 m, the space size of the dataset is 880×1360 pixels, there are 176 bands from 400 to 1000 nm, and the spatial resolution is about 0.15 m, including six types of ground objects, different scales, and containing shadow occlusion.

The TDW dataset was collected from 11:23 to 12:05 on May 18, 2021, using a DJI M600 Pro drone mounted with Gaiaskymini2-VN in Tangdao Bay National Wetland Park, Qingdao, China. The flight height of the UAV is 300 m, the space size of the dataset is 1740×860 pixels, there are 176 bands from 400 to 1000 nm, and the spatial resolution is about 0.15 m, including 18 types of ground objects, which are complex.

Table I provides a detailed overview of the category information in the three drone hyperspectral datasets, including the number of training and testing samples used in the experiment. It is important to note that, to ensure the experiment's validity, the training samples were divided using a regional clustering method rather than a random sampling strategy. This is because random sampling could introduce prior information about geographical location, leading to a misleading improvement in the model's generalization performance. Furthermore, in real-world applications, randomly and independently partitioning each training sample not only requires significant human effort but also lacks practical feasibility. Specifically,

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in this study, the training samples were grouped by clustering 100 samples per region. Additionally, to maintain consistency across experiments, the number of training samples for each category was equal in all datasets, and the overall proportion of training samples across all datasets was kept approximately the same.

B. Network Setting and Evaluation Indexes

All of the experiments are equipped with NVIDIA GeForce RTX 4080 SUPER graphics processor (GPU) unified platform, using PyTorch implementation. The network optimizer uses Adam, and the learning rate is set to 0.01. Using a small batch of size 32, the network is trained for 100 epochs, with the learning rate decaying by one-tenth at epochs 50 and 75. In the two S^{2}DConv branches of LS^{2}DCNet, the convolution kernels of Spatial-DConv are set to five sampling points in an orthogonal shape and three sampling points in a linear shape, respectively; The convolution kernel of Spectral-DConv is uniformly set to three sampling points in a linear shape. In addition, the three adjustable parameters (T, ε, α) of DL-JLoss are set to (1.2, 0.6, 0.3), (1.2, 0.4, 0.6), and (1.2, 0.4, 0.5), respectively, to cope with the LK, QY, and TDW datasets.

This article uses three precision evaluation indices: Kappa coefficient (k), overall accuracy (OA), and average accuracy (AA), and three efficiency evaluation indices: test time, floating-point operations (FLOPs), and number of parameters (Params) to evaluate the performance of LS^{2}DCNet compared with other comparison methods.

Let the confusion matrix be $\mathbf{C} = [c_{ij}]$, where c_{ij} denotes the number of samples whose true class is i and predicted class is j. Suppose there are k classes and the total number of samples is

$$N = \sum_{i=1}^{k}\sum_{j=1}^{k} c_{ij}. \tag{9}$$

The k measures the agreement between predicted and true classifications, accounting for chance agreement

$$\kappa = \frac{N\sum_{i=1}^{k} c_{ii} - \sum_{i=1}^{k}\left(\sum_{j=1}^{k} c_{ij}\right)\left(\sum_{j=1}^{k} c_{ji}\right)}{N^2 - \sum_{i=1}^{k}\left(\sum_{j=1}^{k} c_{ij}\right)\left(\sum_{j=1}^{k} c_{ji}\right)}. \tag{10}$$

OA is the ratio of correctly classified samples to the total number of samples

$$\mathrm{OA} = \frac{\sum_{i=1}^{k} c_{ii}}{N}. \tag{11}$$

AA is the mean of the accuracies for each class

$$\mathrm{AA} = \frac{1}{k}\sum_{i=1}^{k}\frac{c_{ii}}{\sum_{j=1}^{k} c_{ij}}. \tag{12}$$

C. Ablation Experiment

This section first conducted parameter sensitivity experiments on the spatial dimension size of the HSI neighborhood blocks fed into the network, and the experimental results are listed in Table II. The results of the three experimental datasets all show a conclusion: the increase in spatial dimension size does not bring significant accuracy improvement, but increases the cost of time and computing resources. This may be because the original spectrum has already brought enough information to achieve reliable performance using a basic space size. Therefore, according to the results in Table II, the spatial dimension size of HSI neighborhood blocks sent to LS^{2}DCNet is uniformly set to 3×3.

TABLE II
IMPACT OF NEIGHBORHOOD BLOCK SPACE DIMENSION SIZE ON PERFORMANCE

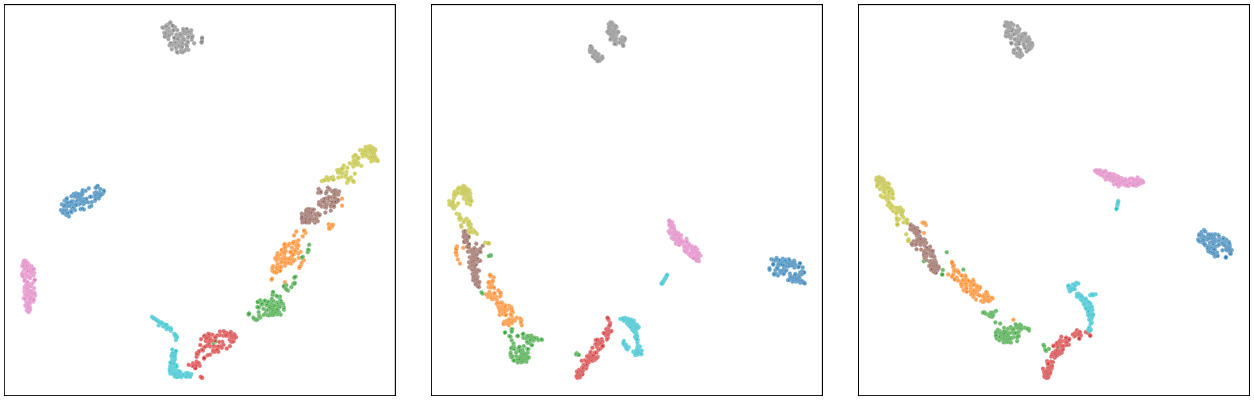
Dataset		3×3	5×5	7×7
LK	$k \times 100$	85.14 ± 1.11	85.39 ± 0.65	**85.66±0.56**
	OA × 100	88.52 ± 0.88	88.73 ± 0.51	**88.95±0.44**
	AA × 100	**81.01±0.78**	80.60 ± 0.71	80.89 ± 0.94
	Test time/s	**2.74±0.10**	4.99 ± 0.09	9.35 ± 1.94
	Params/K	**7.0**	7.1	7.3
	FLOPs/G	**0.1**	0.3	0.7
QY	$k \times 100$	**71.76±1.42**	69.97 ± 1.78	70.96 ± 1.64
	OA × 100	**78.28±1.14**	76.69 ± 1.42	77.67 ± 1.34
	AA × 100	**79.75±0.89**	74.80 ± 0.91	75.35 ± 0.91
	Test time/s	**11.02±0.19**	19.55 ± 5.11	105.55 ± 3.51
	Params/K	**4.1**	4.2	4.4
	FLOPs/G	**0.1**	0.2	0.4
TDW	$k \times 100$	**55.26±0.80**	54.86 ± 1.14	55.13 ± 1.49
	OA × 100	**59.07±0.78**	58.79 ± 1.10	59.03 ± 1.44
	AA × 100	**68.37±0.64**	67.56 ± 0.84	67.78 ± 1.12
	Test time/s	**6.28±0.24**	10.54 ± 0.36	17.21 ± 3.31
	Params/K	**6.2**	6.3	6.5
	FLOPs/G	**0.1**	0.2	0.4

TABLE III
ABLATION EXPERIMENT OF S^{2}DCONV

Dataset		S^{2}DConv	DCNv1	DCNv2	DCNv3	LDConv
LK	$k \times 100$	**85.14±1.11**	83.41 ± 0.58	83.50 ± 0.31	83.01 ± 0.73	84.83 ± 0.48
	OA × 100	**88.52±0.88**	87.15 ± 0.47	87.22 ± 0.25	86.85 ± 0.58	88.30 ± 0.38
	AA × 100	**81.01±0.78**	78.13 ± 0.55	78.21 ± 0.24	77.05 ± 1.04	79.77 ± 0.73
	Test time/s	2.74 ± 0.10	5.44 ± 0.08	7.29 ± 0.17	**2.42±0.10**	3.25 ± 0.10
	Params/K	**7.0**	1403.7	1447.5	174.5	626.1
	FLOPs/G	**0.1**	25.8	26.6	6.3	11.5
QY	$k \times 100$	**71.76±1.42**	66.16 ± 2.46	67.87 ± 1.52	64.46 ± 3.09	68.40 ± 1.27
	OA × 100	**78.28±1.14**	73.65 ± 2.02	75.13 ± 1.32	72.35 ± 2.44	75.73 ± 0.97
	AA × 100	**79.75±0.89**	72.59 ± 1.17	73.18 ± 0.77	71.56 ± 1.94	74.10 ± 0.97
	Test time/s	11.02 ± 0.19	19.32 ± 0.46	25.26 ± 1.09	**10.33±0.20**	13.18 ± 0.49
	Params/K	**4.1**	616.7	645.2	75.3	275.2
	FLOPs/G	**0.1**	11.3	11.8	2.7	5.1
TDW	$k \times 100$	**55.26±0.80**	53.28 ± 0.43	53.48 ± 0.34	55.09 ± 1.67	53.22 ± 1.04
	OA × 100	**59.07±0.78**	57.14 ± 0.41	57.32 ± 0.35	58.96 ± 1.74	56.97 ± 1.05
	AA × 100	**68.37±0.64**	65.60 ± 0.25	65.80 ± 0.30	66.88 ± 1.32	66.90 ± 0.63
	Test time/s	6.28 ± 0.24	11.09 ± 0.16	14.43 ± 0.44	**5.90±0.11**	7.41 ± 0.15
	Params/K	**6.2**	618.8	647.4	77.5	277.4
	FLOPs/G	**0.1**	11.3	11.8	2.7	5.1

Subsequently, the section presents ablation experiments for S^{2}DConv and DL-Jloss to validate their effectiveness. First, the ablation experiment results for S^{2}DConv are listed in Table III. Among them, four popular and advanced dynamic convolutions were selected to replace S^{2}DConv for comparison: DCNv1 [13], DCNv2 [14], DCNv3 [15], and LDConv [17]. In fact, DCNv4 [16] has also been proposed, but it has many limitations, such as the channel size of the input data space being limited to multiples of 16. This will result in greater computational consumption, which is not suitable for the

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(a) (b) (c)

Fig. 4. Feature distribution plots based on T-SNE under different loss functions. (a) CEL. (b) CEL+LS. (c) DL-JLOSS.

TABLE IV
ABLATION EXPERIMENT OF DL-JLOSS

Dataset		DL-JLOSS	CEL	CEL+LS
LK	$k \times 100$	**85.14±1.11**	84.53 ± 0.77	82.35 ± 1.13
	OA × 100	**88.52±0.88**	88.05 ± 0.60	86.32 ± 0.91
	AA × 100	**81.01±0.78**	80.48 ± 1.34	76.39 ± 0.64
	Test time/s	2.74 ± 0.10	2.63 ± 0.07	**2.59±0.09**
	Params/K	7.0	7.0	7.0
	FLOPs/G	0.1	0.1	0.1
QY	$k \times 100$	**71.76±1.42**	70.64 ± 1.18	70.12 ± 1.21
	OA × 100	**78.28±1.14**	77.40 ± 0.91	77.04 ± 0.97
	AA × 100	**75.95±0.89**	75.09 ± 0.80	74.86 ± 0.63
	Test time/s	11.02 ± 0.19	11.03 ± 0.17	**10.96±0.25**
	Params/K	4.1	4.1	4.1
	FLOPs/G	0.1	0.1	0.1
TDW	$k \times 100$	**55.26±0.80**	54.86 ± 1.13	52.12 ± 0.65
	OA × 100	**59.07±0.78**	58.68 ± 1.13	56.10 ± 0.66
	AA × 100	**68.37±0.64**	67.95 ± 0.46	64.70 ± 0.56
	Test time/s	6.28 ± 0.24	6.62 ± 0.36	**6.22±0.13**
	Params/K	6.2	6.2	6.2
	FLOPs/G	0.1	0.1	0.1

UAV HSI classification task targeted in this article. Therefore, DCNv4 [16] was not compared here. From Table III, it can be seen that S^2DConv achieved the best classification accuracy and the minimum number of parameters and computational complexity on three datasets. In terms of computation time, DCNv3 [15] achieved the best performance, but its difference from S^2DConv is only an average of no more than 0.5 s. Therefore, based on various indicators, S^2DConv demonstrates its superiority in UAV HSI classification tasks.

The results of ablation experiments for DL-JLoss are listed in Table IV. The comparison method selected a single CEL and the combination of CEL and LS to verify the effectiveness of DL-Jloss. As can be seen from Table IV, compared with CEL alone, the combination of CEL and LS both resulted in a decrease in classification accuracy to varying degrees, while the introduction of DL-JLoss improved classification accuracy to varying degrees on all three datasets, which verified the effectiveness of DL-JLoss. In this case, the negative gain of LS on classification accuracy may be because it ignores the distribution relationship between different categories in HSI, and the operation of "equalization" leads the network to learn wrong information. In addition, compared to CEL alone, the accuracy of DL-JLoss is improved but not significantly. On the one hand, this may be due to the small number of training samples and the high difficulty of the task. On the other hand, it may be because the adaptive extraction of ground features by S^2DConv itself can partially improve the network generalization performance, so the gain of using DL-JLoss is not outstanding. In addition, Fig. 5 displays the visualization of feature distributions under different loss functions using T-SNE. The results more clearly demonstrate that DL-JLoss further reduces intraclass distances while creating more distinct and accurate interclass distribution structures.

D. Classification Results and Discussion

Whether it is UAV HSI or satellite HSI, labeling training samples is a task with high labor costs. In particular, UAV HSI has a higher spatial resolution, which leads to a higher cost of labeling training samples. Therefore, we should pay attention to the research of high-performance UAV HSI classification algorithms with a small number of training samples. In this article, a small number of training samples with roughly the same proportion were set on the three experimental datasets for training each algorithm, as listed in Table I. The performance comparison results on the three experimental datasets are listed in Table V. The comparison method selects the existing popular and most advanced algorithms, including F3D [35] based on 3-D convolution and GloRe_HCNN [36] based on hybrid convolution, and lightweight networks: CAN [7], LIRnet [37], LiteDepthwiseNet [25], LSSAN [26], and Lite-HCNet [27].

LK, QY, and TDW are UAV HSI datasets of three different scenarios, namely, farmland, architectural buildings, and park areas, which can well measure the generalization performance of the UAV HSI classification algorithm for different scenarios. As can be seen from Table V, combining classification accuracy and efficiency indicators, LS^2DCNet achieved the best performance with a small number of training samples on the three experimental datasets. In terms of classification accuracy, LS^2DCNet achieves the highest accuracy on the

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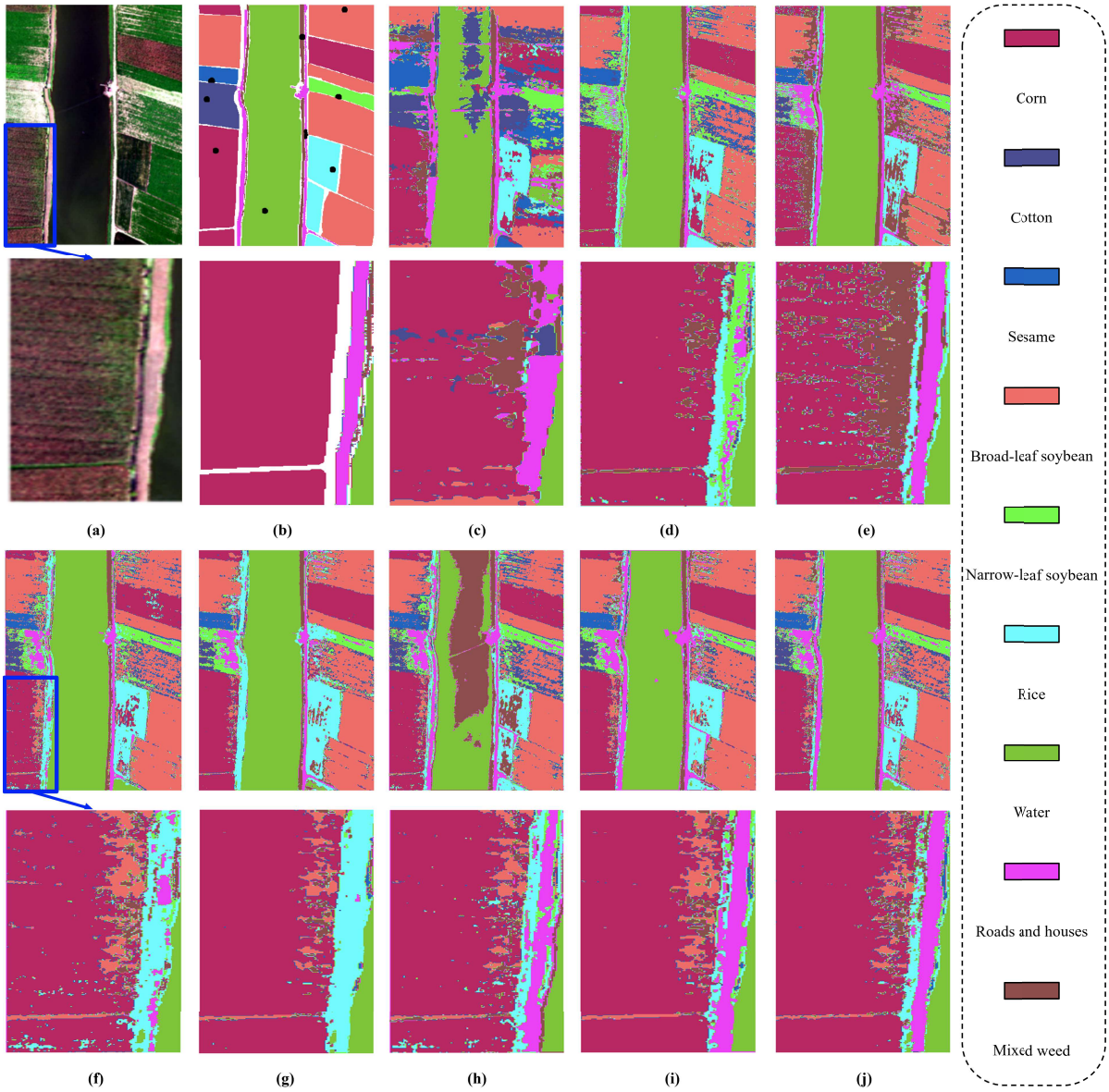


Fig. 5. Classification maps of different methods on LK. (a) Pseudo RGB image. (b) Ground truth and distribution of training samples. (c) CAN (AA: 61.09%). (d) LIRnet (AA: 72.14%). (e) GloRe_HCNN (AA: 72.53%). (f) F3D (AA: 72.15%). (g) LiteDepthwiseNet (AA: 70.00%). (h) LSSAN (AA: 66.56%). (i) Lite-HCNet (AA: 77.61%). (j) LS^{2}DCNet (AA: 79.47%).

three experimental datasets. Other comparison methods that achieve the best on some metrics, such as CAN, which achieves the best AA metrics on the QY dataset and TDW dataset, show no equivalence on other metrics. This also verifies that LS^{2}DCNet has better generalization performance and can deal with complex and changeable scenes. In terms of efficiency, LS^{2}DCNet has demonstrated significant advantages, not only with fewer parameters and calculations but also with less testing time. Taking Lite-HCNet with similar classification accuracy as examples, the test time on the LK dataset is 4.6 times that of LS^{2}DCNet, and the parameter number is 66 times that of LS^{2}DCNet, respectively. The computational capacity is 98 times that of LS^{2}DCNet, respectively. It is worth noting that although LSSAN achieves the least test time, its classification accuracy is much lower than that of LS^{2}DCNet, and its comprehensive performance is not good. In summary, LS^{2}DCNet can achieve higher classification accuracy with fewer computing resources and higher operating efficiency under a small number of training samples, which is suitable for UAV HSI classification scenarios with limited computing resources and high time requirements.

Figs. 5–7, respectively, show the comparison of the visual classification results of the overall and local details of each algorithm on the three experimental datasets. Among the ten statistical results of each algorithm, the lowest AA index is selected to demonstrate the stability of the algorithm.

First, it can be seen that in addition to the QY dataset, LS^{2}DCNet achieves the best performance for the lowest AA index in the results of ten runs on the LK and TDW datasets, and the gap between the average results and other methods is smaller, which proves its stability. In addition, the visual classification effect of LS^{2}DCNet is generally good,

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TABLE V
COMPLEXITY AND CLASSIFICATION ACCURACIES OF DIFFERENT MODELS ON EXPERIMENTAL DATASETS

Dataset		CAN [7]	LIRnet [37]	GloRe_HCNN [36]	F3D [35]	LiteDepthwiseNet [25]	LSSAN [26]	Lite-HCNet [27]	LS^{2}DCNet
LK	$k \times 100$	58.33 ± 2.42	84.31 ± 2.83	83.81 ± 3.66	81.36 ± 1.11	83.84 ± 1.52	79.77 ± 5.58	84.00 ± 0.54	**85.14±1.11**
	OA × 100	65.58 ± 2.20	87.87 ± 2.27	87.45 ± 3.01	85.60 ± 0.88	87.59 ± 1.18	84.18 ± 4.81	87.67 ± 0.42	**88.52±0.88**
	AA × 100	73.75 ± 1.33	79.03 ± 3.56	78.93 ± 2.57	75.08 ± 1.51	74.87 ± 2.81	73.57 ± 2.58	80.10 ± 1.05	**81.01±0.78**
	Test time/s	5.31 ± 0.19	4.63 ± 0.09	6.78 ± 0.04	6.97 ± 0.86	5.14 ± 0.04	**2.09±0.07**	12.71 ± 0.12	2.74 ± 0.10
	Params/K	338.3	71.8	23734.4	37858.5	66.4	726.2	461.3	**7.0**
	FLOPs/G	3.8	1.3	305.9	434.1	15.1	12.9	9.8	**0.1**
QY	$k \times 100$	71.21 ± 2.32	67.05 ± 1.44	71.56 ± 2.04	57.61 ± 2.23	68.75 ± 3.27	56.93 ± 2.96	71.24 ± 1.37	**71.76±1.42**
	OA × 100	77.50 ± 1.88	74.60 ± 1.14	78.08 ± 1.59	66.83 ± 1.98	75.65 ± 2.66	66.48 ± 2.28	77.93 ± 1.08	**78.28±1.14**
	AA × 100	**79.11±4.80**	73.10 ± 0.65	76.54 ± 1.28	67.99 ± 1.18	74.92 ± 1.67	67.87 ± 1.65	76.14 ± 0.82	75.95 ± 0.89
	Test time/s	24.80 ± 1.63	16.24 ± 4.84	21.98 ± 0.25	19.23 ± 0.13	16.80 ± 0.31	**8.94±0.25**	37.02 ± 0.29	11.02 ± 0.19
	Params/K	277.4	65.7	11238.8	23997.3	45.9	336.7	201.0	**4.1**
	FLOPs/G	2.6	1.2	151.7	275.1	9.7	5.7	4.5	**0.1**
TDW	$k \times 100$	53.74 ± 1.22	53.80 ± 1.89	54.17 ± 6.40	52.20 ± 2.92	38.35 ± 6.39	49.89 ± 1.86	51.75 ± 2.80	**55.26±0.80**
	OA × 100	57.35 ± 1.26	57.74 ± 1.81	57.93 ± 6.42	56.41 ± 3.02	43.14 ± 6.87	54.21 ± 1.79	55.77 ± 2.83	**59.07±0.78**
	AA × 100	**73.07 ± 1.42**	64.17 ± 1.81	65.66 ± 6.98	58.73 ± 1.90	43.80 ± 4.25	57.20 ± 1.24	64.40 ± 1.19	68.37 ± 0.64
	Test time/s	22.55 ± 6.14	5.58 ± 0.27	12.64 ± 0.14	11.09 ± 0.04	9.77 ± 0.14	**5.03±0.11**	22.34 ± 0.12	6.28 ± 0.24
	Params/K	277.4	66.1	11240.3	23998.8	46.6	337.1	203.1	**6.2**
	FLOPs/G	2.6	1.2	151.7	275.1	9.7	5.7	4.5	**0.1**

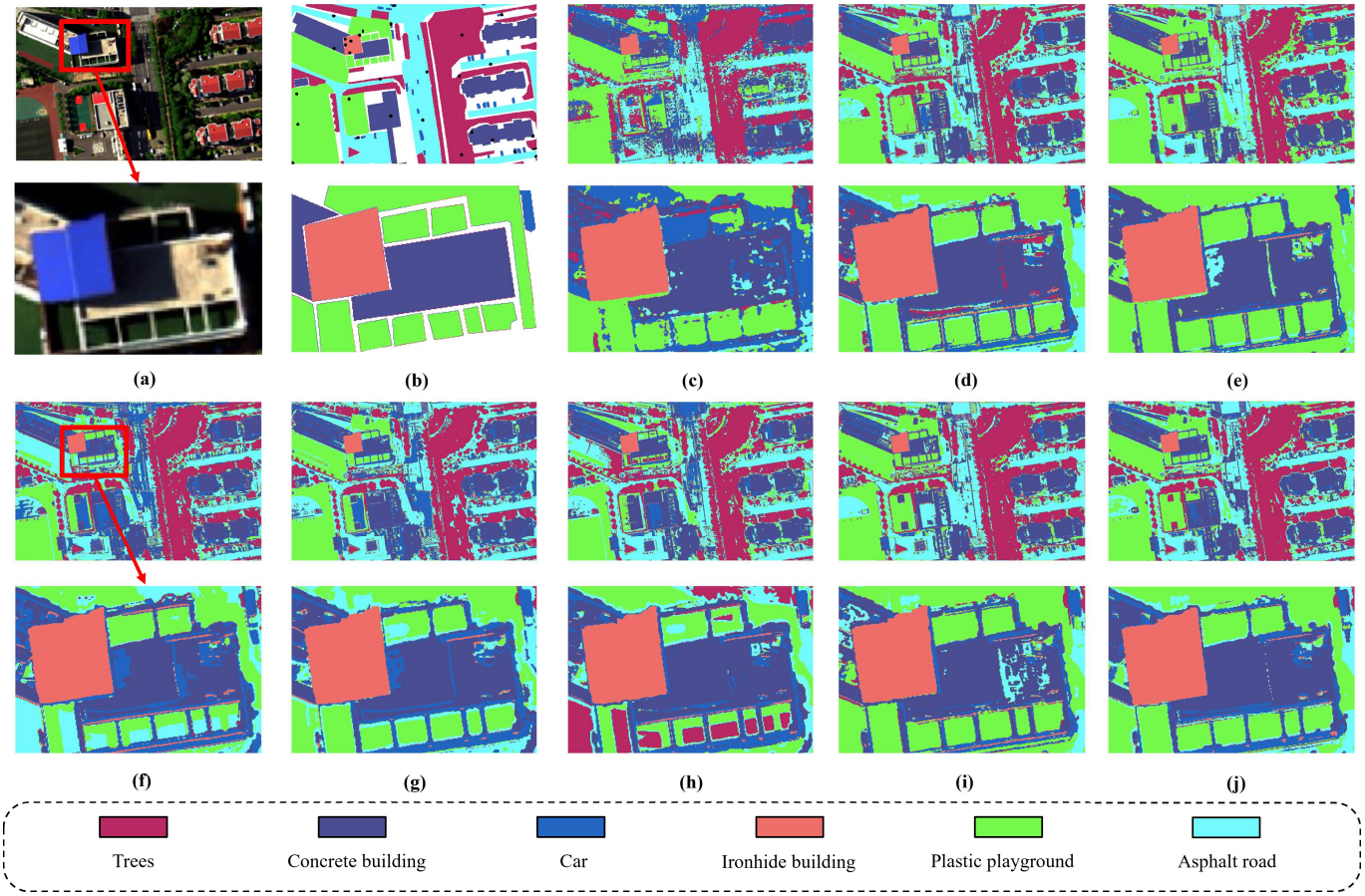


Fig. 6. Classification maps of different methods on QY. (a) Pseudo RGB image. (b) Ground truth and distribution of training samples. (c) CAN (AA: 65.40%). (d) LIRnet (AA: 72.38%). (e) GloRe_HCNN (AA: 74.31%). (f) F3D (AA: 65.29%). (g) LiteDepthwiseNet (AA: 72.77%). (h) LSSAN (AA: 66.28%). (i) Lite-HCNet (AA: 74.59%). (j) LS^{2}DCNet (AA: 73.83%).

and the local details highlight its advanced nature. On the LK dataset, compared to other methods, LS^{2}DCNet not only generates a relatively smoother distribution of corn but also has clearer boundaries for roads and houses. On the QY dataset, compared with other methods, the plastic playgrounds and buildings generated by LS^{2}DCNet have relatively more complete internal distribution and clearer boundaries. On TDW datasets, LS^{2}DCNet also achieves relatively clearer boundaries than other methods. This is due to the high dynamic adaptive extraction capability of the designed S^{2}DConv for detailed terrain distribution, such as boundaries. And the designed DL-JLoss has improved the network's generalization performance.

Fig. 8 illustrates the trend in classification performance as the number of training samples gradually increases on

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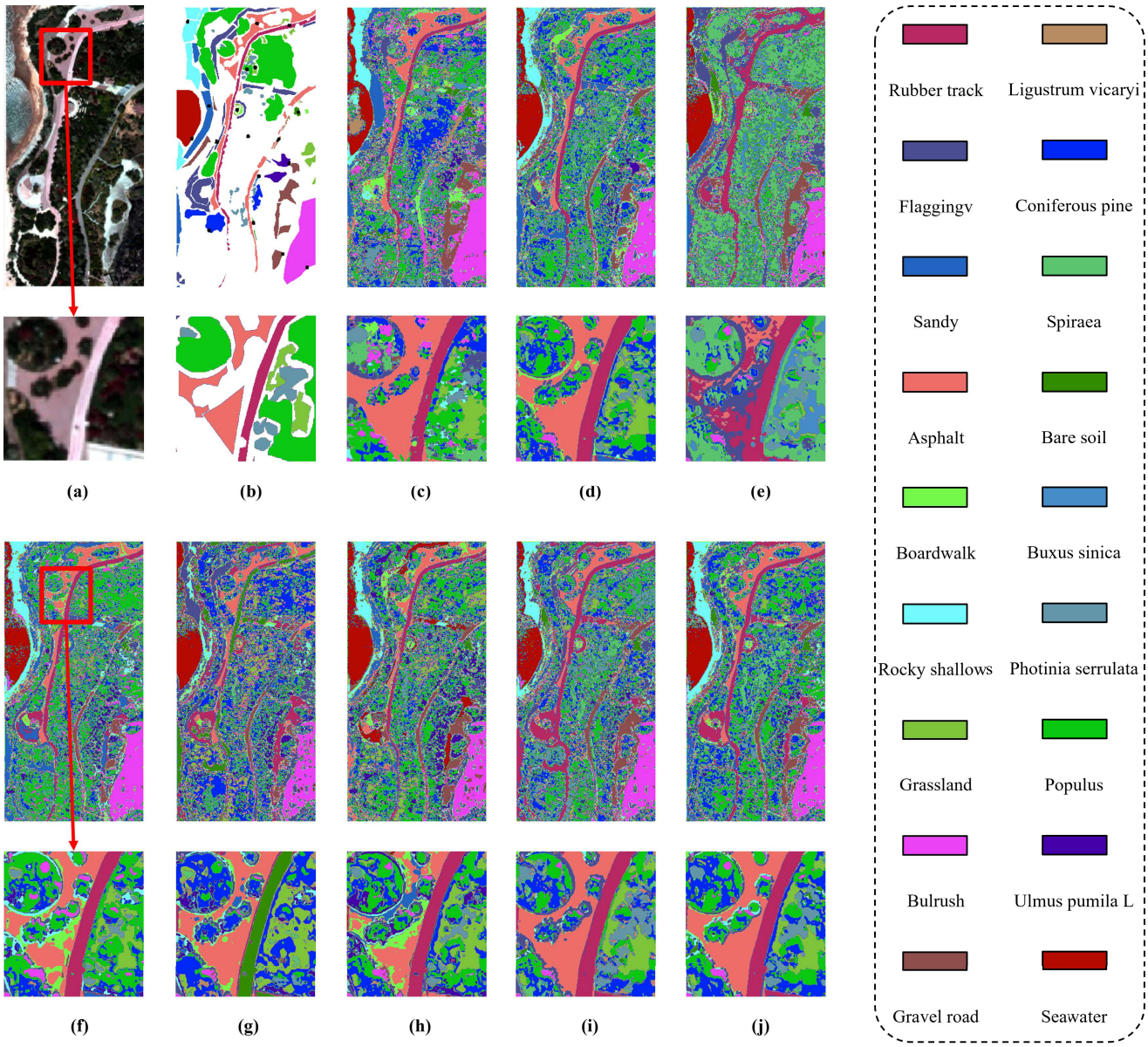


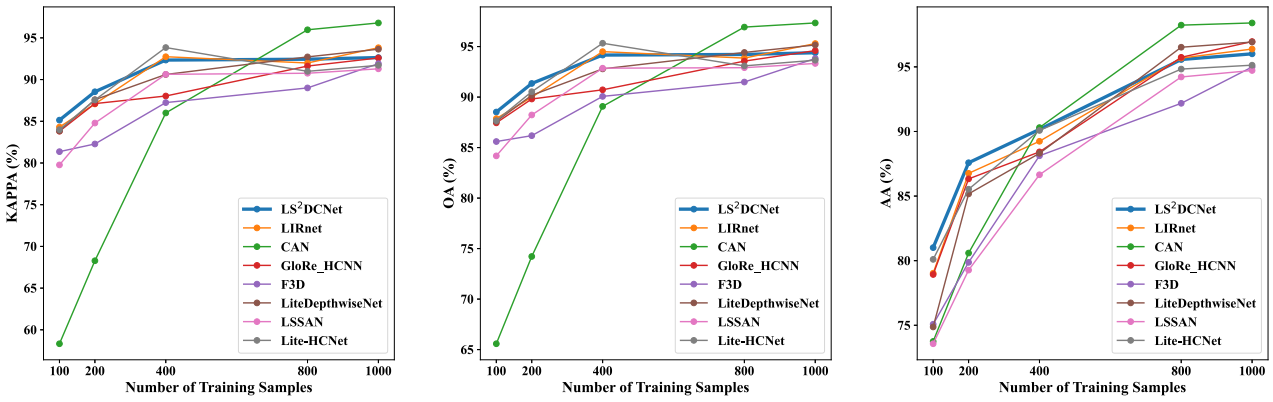
Fig. 7. Classification maps of different methods on TDW. (a) Pseudo RGB image. (b) Ground truth and distribution of training samples. (c) CAN (AA: 70.76%). (d) LIRnet (AA: 61.24%). (e) GloRe_HCNN (AA: 45.96%). (f) F3D (AA: 55.46%). (g) LiteDepthwiseNet (AA: 38.24%). (h) LSSAN (AA: 55.42%). (i) Lite-HCNet (AA: 62.39%). (j) LS^{2}DCNet (AA: 67.24%).

the LK dataset, focusing on three key metrics: k, OA, and AA. Specifically, 100, 200, 400, 800, and 1,000 training samples were selected for each feature type, with samples divided according to the principle of aggregating 100 samples per geographical area. In other words, when each feature type reaches 1000 training samples, they cover ten different geographical regions. This change also provides more geographical prior information, reducing the difficulty of testing the model's generalization performance. The results show that as the number of training samples grows, all compared models exhibit a significant positive correlation, confirming the link between sample size and classification performance. Notably, the CAN model performs poorly with small samples, but as the training sample size increases, its performance curve rises steeply, eventually surpassing other models. This indicates the strong dependence of the CAN architecture on large-scale geographical prior data and highlights its potential in scenarios with sufficient training resources. Conversely, the LS^{2}DCNet proposed in this article shows robust performance advantages under limited training samples and geographical prior data, maintaining good performance even as the dataset expands. Without considering manual data collection and annotation costs, CAN may be a very competitive choice. However, in scenarios with limited training samples, restricted geographical prior data, and limited computational resources, LS^{2}DCNet is undoubtedly a highly practical and effective solution.

E. Embedded Device Inferencing Experiment

To verify the real situation of LS^{2}DCNet in the UAV HSI classification task with limited computing resources, this

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(a) (b) (c)

Fig. 8. Metric variation curves under different numbers of training samples. (a) KAPPA(%). (b) OA(%). (c) AA(%).

TABLE VI
COMPARISON OF INFERENCE TIME ON EMBEDDED DEVICE

Method		LK	QY	TDW
CAN [7]	Inference time/s	16	50	71
LIRnet [37]	Inference time/s	8	26	50
GloRe_HCNN [36]	Inference time/s	80	172	224
F3D [35]	Inference time/s	37	116	172
LiteDepthwiseNet [25]	Inference time/s	27	86	120
LSSAN [26]	Inference time/s	**7**	**20**	**26**
Lite-HCNet [27]	Inference time/s	29	97	188
LS^{2}DCNet	Inference time/s	10	32	55

article sets up a reasoning experiment using the PyTorch deep learning framework on the NVIDIA Jetson AGX Orin 64GB embedded device. NVIDIA Jetson AGX Orin 64 GB has 275 TOPS AI performance. Table VI shows the inference time comparison between LS^{2}DCNet and other comparison methods on embedded devices. It can be seen that the results in Table VI follow the rules in Table V. Among them, LSSAN achieved the lowest time consumption, but its accuracy metrics were not ideal. In contrast, LF-S^{2}DCNet demonstrated competitive results, and when considering accuracy metrics, its overall performance was better.

Although LS^{2}DCNet has fewer parameters and lower computational complexity, it does not have a significant advantage in runtime. This may be due to tensor concatenation and transformation operations in the model, which cause fragmented memory access patterns and affect actual runtime efficiency. To address this issue, future optimization efforts should focus on reducing memory access overhead to improve the model's runtime speed and resource use.

IV. CONCLUSION

In this article, an LS^{2}DCNet is designed for UAV HSI classification. In LS^{2}DCNet, S^{2}DConv is designed to enhance the adaptive ability of the network to extract fine features at high spatial resolution. DL-JLoss is designed to dynamically learn the distribution relationship between classes under complex ground objects to improve the generalization performance of the network. Experiments on three UAV HSI datasets validate the comprehensive performance advantages of LS^{2}DCNet in terms of classification accuracy and efficiency. The algorithm proposed in this article, designed for pixel-by-pixel classification tasks, is more suitable for small-scale ground object classification. Large-scale ground object classification is more effectively addressed using segmentation networks. Therefore, in future work, we will extend the idea of this work to segmentation tasks based on UAV HSI.

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